

Abstract No.

CD9 protein in glioblastoma: Implications for progression and therapeutic strategies.

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Glioblastoma (GB) contains a population of glioma stem like cells (GSCs) that resemble adult neural stem cells (NSCs) and contribute to treatment resistance and tumour relapse. CD9 (Tspan29) is a tetraspanin enriched in NSCs in the ventricular–subventricular zone and striatum, where it is linked to a neurogenic like programme, and it is also increased in GBM, where higher levels associate with invasion, resistance to cell death and worse prognosis. In this project, we study the role of CD9 in maintaining a stem like state and in regulating communication between glioma cells and their microenvironment using a murine glioblastoma stem like cell model. We use genetic and cellular approaches to analyse how changes in CD9 affect cell proliferation, stemness and differentiation markers, both at baseline and after chemotherapy.

By combining these results with published data on CD9 in neural stem cells and GBM, we aim to understand whether CD9 helps connect a neural stem like programme with signals that support tumour growth.

This work may help explain how a normal brain stem cell marker is reused by GBM stem like cells and point to new ways to interfere with tumour–microenvironment communication to improve treatment response.



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